

ProjectInitializationandPlanningPhase

Date	1July2026
TeamID	XXXXXX
ProjectTitle	EduGenie – Google Gemini Powered Learning Assistant
MaximumMarks	5Marks

ProjectProposal(ProposedSolution)Report

The proposed solution introduces EduGenie – Google Gemini Powered Learning Assistant, an AI-powered educational platform that enhances learning through intelligent academic assistance. The system uses Google Gemini 1.5 Pro and LaMini-Flan-T5 models to provide concept explanations, answer academic questions, generate quizzes, summarize educational content, and recommend personalized learning paths. The application delivers accurate, interactive, and personalized learning support through an easy-to-use web interface.

Project Overview

Topic	Description
Objective	Develop an intelligent AI-powered educational assistant that helps students understand concepts, answer academic questions, generate quizzes, summarize learning materials, and provide personalized learning recommendations.
Scope	The system supports concept explanation, question answering, quiz generation, text summarization, personalized learning paths, AI-based educational assistance, and a responsive web interface for students and educators.

Problem Statement

Description	Students often struggle to understand complex concepts, search across multiple resources, create study materials, and receive personalized guidance for effective learning. Existing platforms provide fragmented educational services instead of an integrated AI learning solution.
Impact	An AI-powered educational assistant improves learning efficiency, reduces study time, enhances conceptual understanding, and provides personalized academic support for students.

Proposed Solution

Topic	Details
-------	---------

Topic	Description
-------	-------------

Approac h	Use Google Gemini 1.5 Pro, LaMini-Flan-T5, FastAPI, and modern web technologies to process user requests and generate AI-powered educational content, including explanations, answers, quizzes, summaries, and personalized learning recommendations.
Key Features	• Concept Explanation • Question Answering (QnA) • Quiz Generation • Text Summarization • Personalized Learning Path Recommendations • AI-powered Educational Assistance • Responsive Web Interface

ResourceRequirements

ResourceType	Description	Specification/Allocation
Hardware		
Processor	CPU	IntelCorei5/Ryzen5orhigher
Memory	RAM	8GBorabove
Storage	DiskSpace	10GBfreeSSDspace
Software		
Framework	Application Framework	FastAPI
Libraries	PythonLibraries	Google Generative AI SDK, Transformers, Pydantic, Jinja2, Uvicorn
Development Environment	IDE	VisualStudioCode

Data		
Dataset	ReferenceConcepts& Audio Samples	Learning topics, user queries, quizzes, summaries, and personalized learning recommendations